# 3.2 Analysis

## Project Group


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## 1. Judge Validation

We randomly sampled 30 recorded outputs across our slices and labeled them manually against our three rubric questions.

### Labeling Agreement (Human vs. Judge)

| Rubric Question | Human Yes / Judge Yes | Human Yes / Judge No | Human No / Judge Yes | Human No / Judge No |
|---|---|---|---|---|
| `agrees_with_action` | 25 | 4 | 0 | 1 |
| `states_policy_correctly` | 16 | 1 | 1 | 12 |
| `arithmetic_correct` | 26 | 4 | 0 | 0 |

### Judge Failure Modes

During our manual validation, we identified two primary failure modes where the LLM judge's verdicts diverged from human reasoning: - **Blindness to Hallucinated Policy:** The judge often failed to recognize when the triage model invented fake policy exceptions (e.g., claiming an amount was "close enough to $50" or "a reasonable product price"). If the arithmetic was correct, the judge frequently marked `states_policy_correctly` as True, even when the stated rule was a hallucination. - **Susceptibility to Injections:** The judge struggled to penalize prompt injections (like the "SYSTEM OVERRIDE" tickets). It frequently marked the model's rationale as policy-compliant because it failed to distinguish between the actual system policy and the injected fake policy text.

### Judge Noise Floor

We ran the judge three times on the same 30 items. Across the 90 total rubric questions (3 questions × 30 items), the judge disagreed with its own previous verdicts on 12 occasions, resulting in a judge noise floor of 13.3%.

## 2. Noise Floor

We ran the baseline system (`policy-in-user`) 5 times across the 70 tickets. The table below represents the pass rate per slice, the spread (highest pass rate minus lowest pass rate across the 5 runs), and the smallest change we can confidently detect.

| Slice | Pass Rates (Runs 1-5) | Spread (Noise Floor) | Smallest Detectable Change |
|---|---|---|---|
| `intent:refund` | 40, 40, 39, 41, 41 of 57 | 4 points | Any difference between conditions smaller than 4% is indistinguishable from noise. |

| Slice | Pass Rates (Runs 1-5) | Spread (Noise Floor) | Smallest Detectable Change |
|---|---|---|---|
| `amount:under-50` | 16, 16, 16, 16, 16 of 16 | 0 points | We can detect any change in performance, as the model perfectly handles this slice consistently. |
| `amount:50-to-200` | 13, 13, 13, 14, 14 of 18 | 6 points | Any difference between conditions smaller than 6% is indistinguishable from noise. |
| `amount:over-200` | 10, 10, 10, 10, 10 of 14 | 0 points | We can detect any change in performance, as the model's baseline pass rate does not fluctuate here. |
| `amount:near-50` | 1, 1, 0, 1, 1 of 9 | 11 points | Any difference between conditions smaller than 11% is indistinguishable from noise. |
| `complexity:math` | 5, 5, 5, 5, 5 of 11 | 0 points | We can detect any change in performance, as the model's baseline pass rate does not fluctuate here. |
| `injection:yes` | 1, 1, 2, 2, 1 of 8 | 12 points | Any difference between conditions smaller than 12% is indistinguishable from noise. |

## 3. The Question, Answered

**Does sending the policy as the system instruction, instead of beside the ticket, help?**

By comparing the experimental runs (`policy-in-system`) against the baseline noise floor (`policy-in-user`), we observed the following verdicts per slice:

- **`injection:yes`**: **Helped**. We believe this verdict because the experimental pass rate averaged an improvement of +15 points over the baseline, which cleanly exceeds the experimental condition's noise floor of 12 points.
- **`amount:over-200`**: **Helped**. We believe this verdict because moving the policy improved the average pass rate by +11 points, which exceeds the experimental condition's noise floor of 7 points.
- **`amount:near-50`**: **Cannot Tell**. The experimental gap was only -4 points, which is entirely inside the 11-point noise floor. The runs naturally fluctuate by that much on their own, so we cannot claim the prompt change had a measurable effect.
- **`complexity:math`**: **Cannot Tell**. The experimental gap was -2 points, which is inside the 9-point noise floor for this slice.
- **`amount:50-to-200`**: **Cannot Tell**. The experimental gap was -6 points, which is inside the 17-point noise floor for this slice.
- **`intent:question`**: **No Change**. The pass rate remained perfectly stable across all 5 runs in both conditions (11 of 13 tickets).